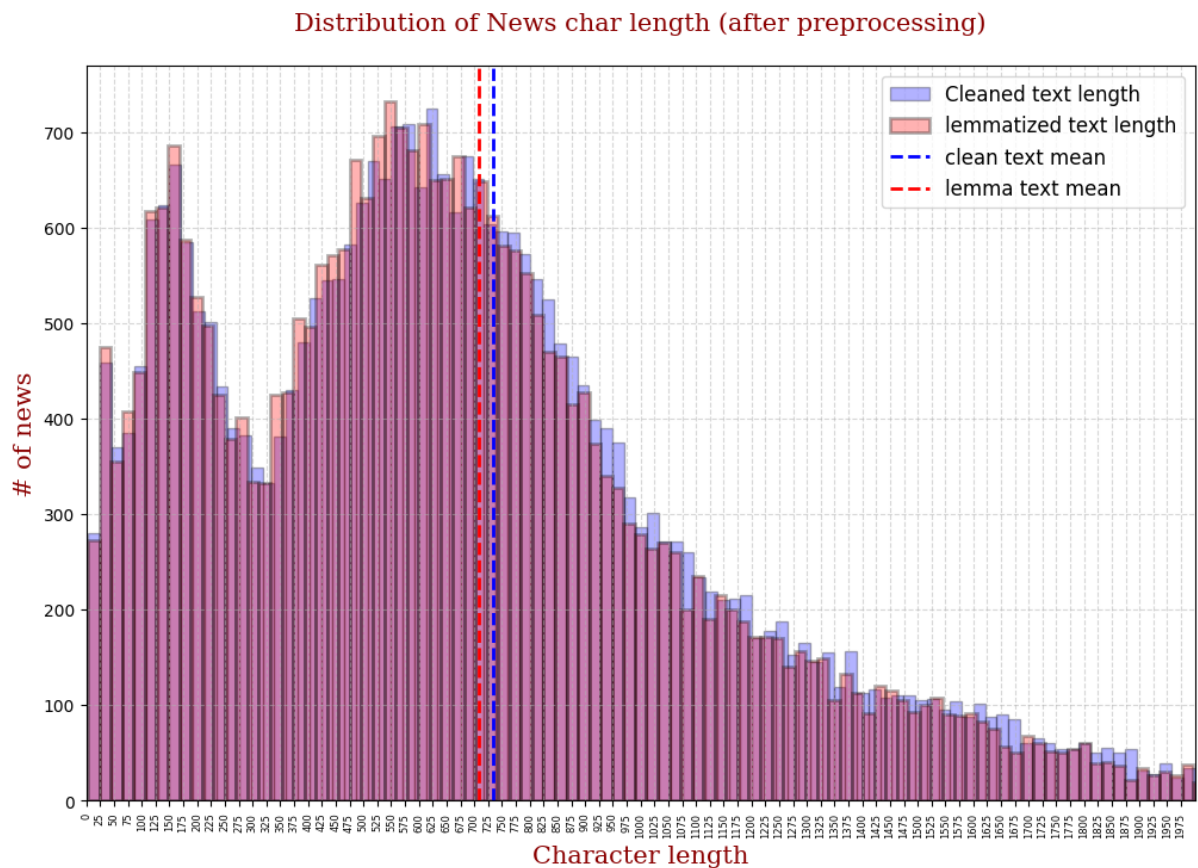


4.1 Visualise character lengths of cleaned news text and lemmatized news text with POS tags removed

4.1.2 Create Histogram to visualise character lengths [7 marks]

Plot both distributions on the same graph for comparison and to observe overlaps and peak differences to understand text preprocessing's impact on text length.



4.2 Find and display the top 40 words by frequency among true and fake news in Training data after processing the text

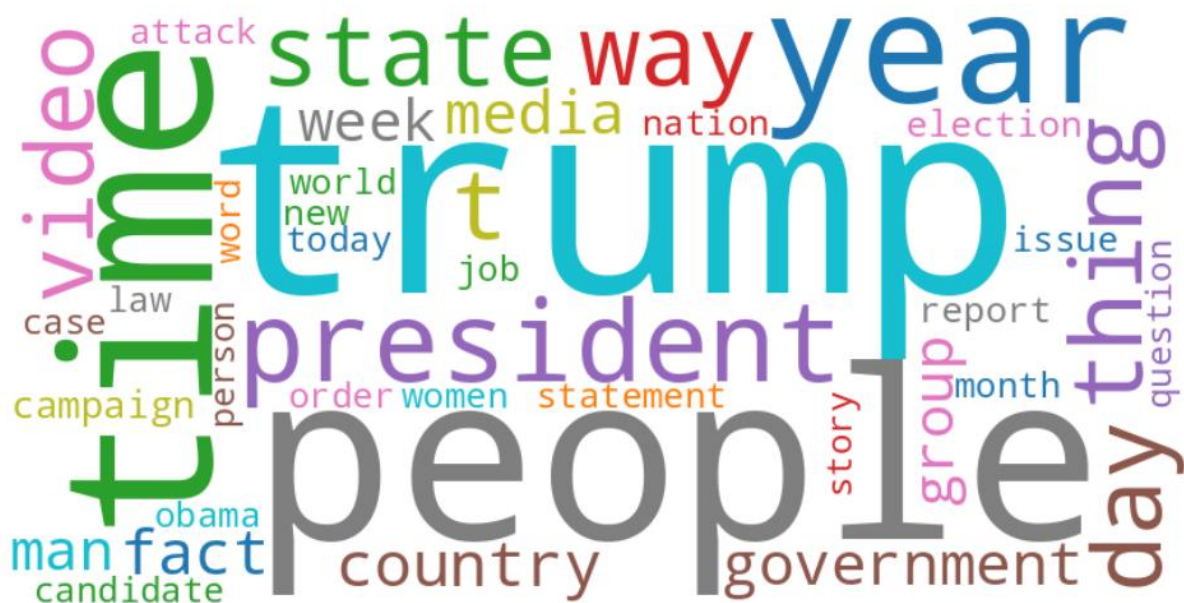
4.2.1 Find and display the top 40 words by frequency among true news in Training data after processing the text

Top 40 Words in True News (After Preprocessing)



4.2.2 Find and display the top 40 words by frequency among fake news in Training data after processing the text

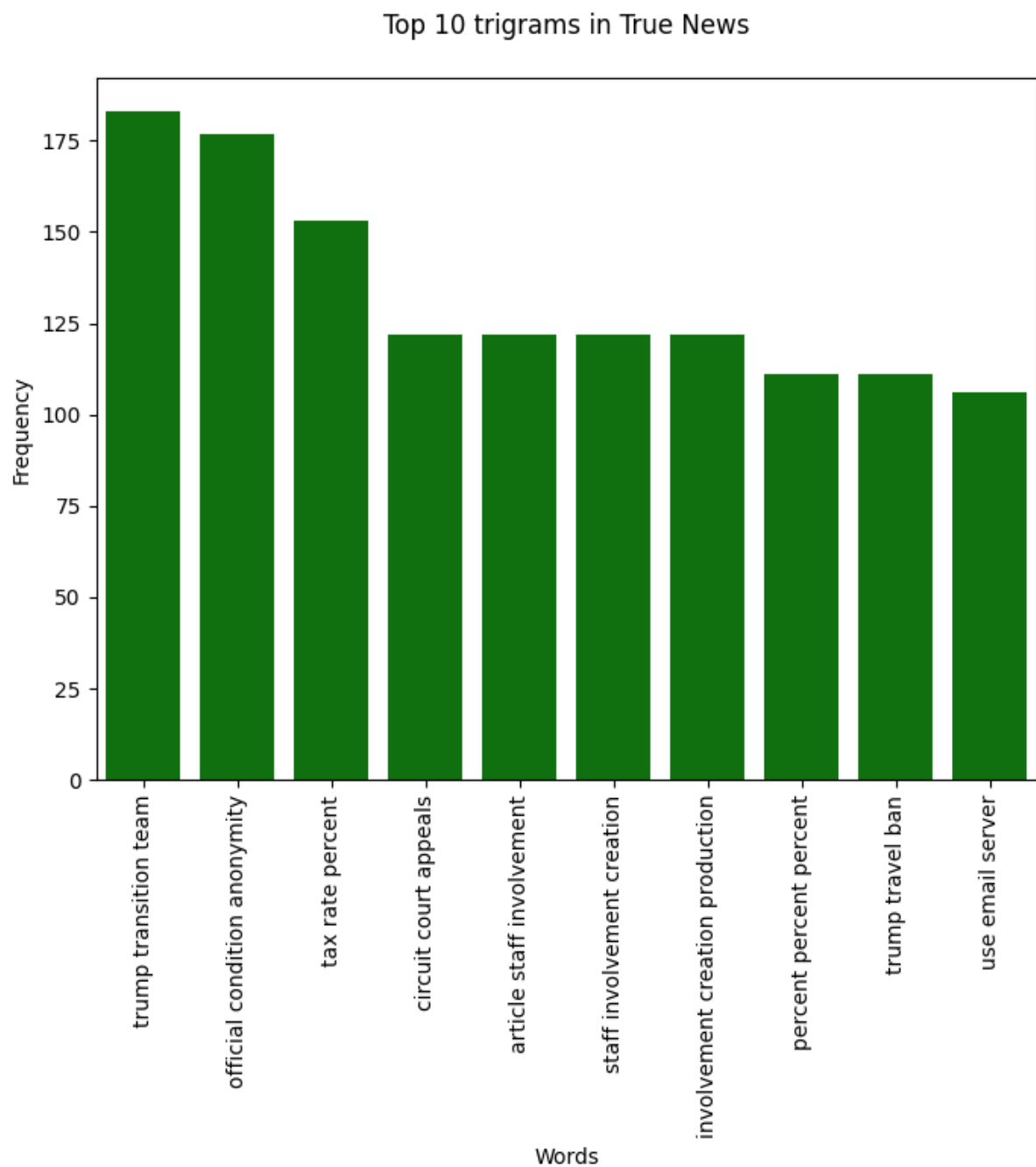
Top 40 Words in Fake News (After Preprocessing)



4.3.5 Display the top 10 trigrams by frequency in true news and plot them as a bar graph

Top 10 trigrams in True news are [('trump transition team', 183), ('official condition anonymity', 177), ('tax rate percent', 153), ('circuit court appeals', 122), ('article staff involvement', 122), ('staff

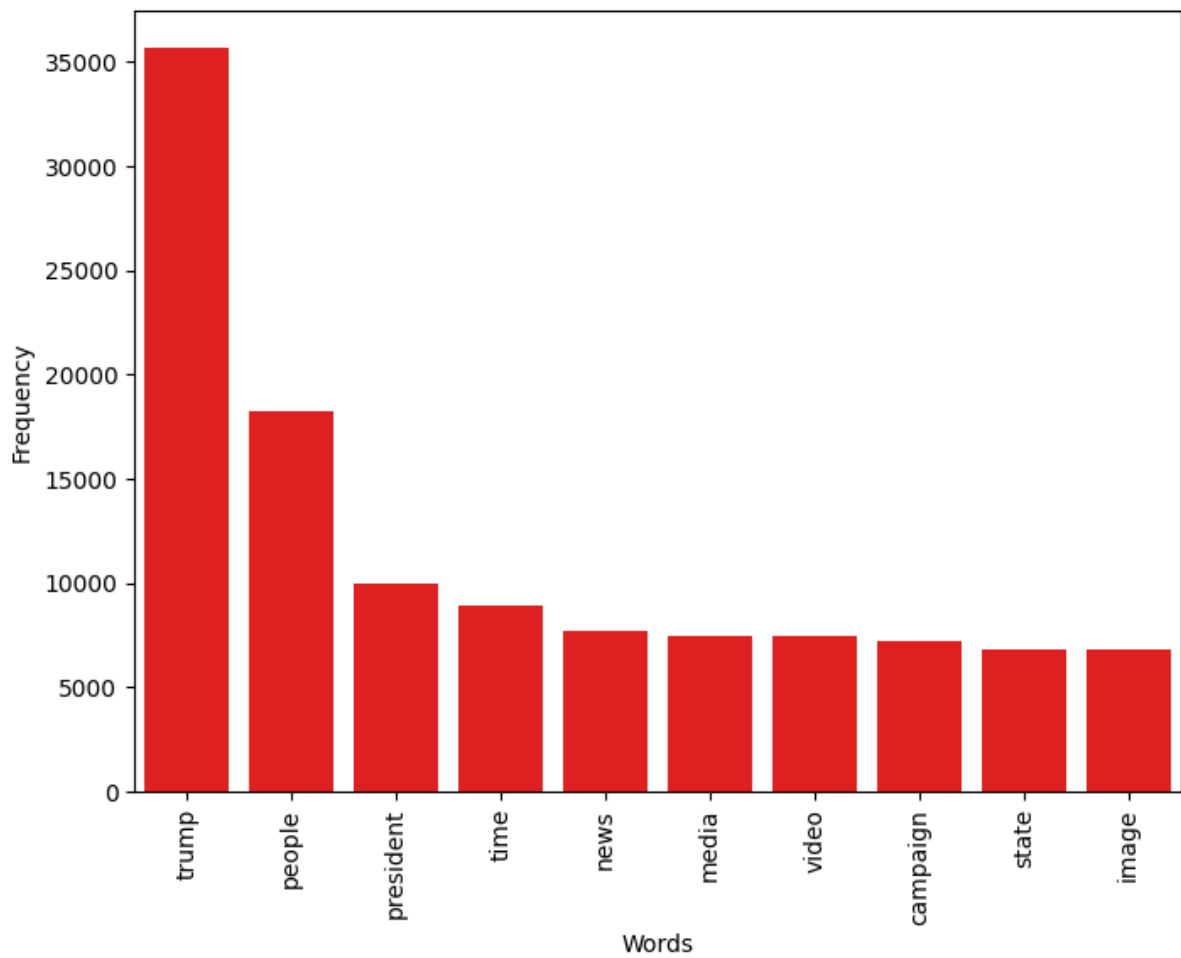
involvement creation', 122), ('involvement creation production', 122), ('percent percent percent', 111), ('trump travel ban', 111), ('use email server', 106)]



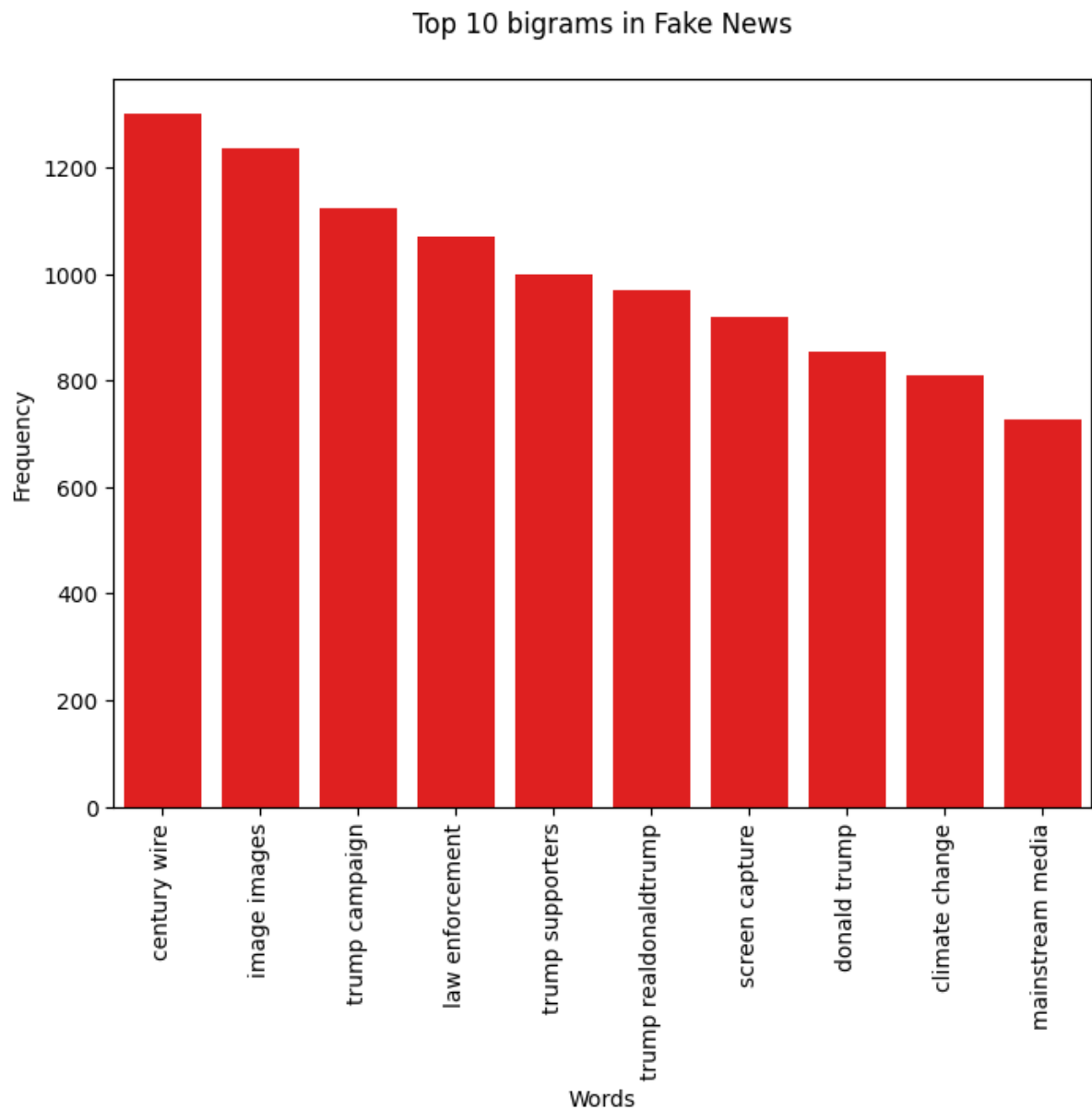
4.3.6 Display the top 10 unigrams by frequency in fake news and plot them as a bar graph

Top 10 unigrams in fake news are [('trump', 35676), ('people', 18209), ('president', 9997), ('time', 8937), ('news', 7684), ('media', 7497), ('video', 7417), ('campaign', 7247), ('state', 6790), ('image', 6785)]

Top 10 unigrams in fake News



Top 10 bigrams in Fake news are [('century wire', 1302), ('image images', 1236), ('trump campaign', 1122), ('law enforcement', 1069), ('trump supporters', 1000), ('trump realdonaldtrump', 969), ('screen capture', 919), ('donald trump', 853), ('climate change', 811), ('mainstream media', 727)]



Top 10 trigrams in Fake news are [('news century wire', 604), ('video screen capture', 516), ('image video screen', 504), ('image getty images', 395), ('image screen capture', 306), ('image chip images', 179), ('century wire files', 166), ('image video screenshot', 162), ('broadcast boiler room', 156), ('tune radio network', 146)]

Top 10 trigrams in False News

